The Impact of Three-Point Shooting on NBA Team Success

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Spring 2025





Introduction

Motivation: The modern NBA looks radically different from even a decade ago, largely due to the rise of the three-point shot. What was once seen as a situational strategy has evolved into a fundamental pillar of basketball, with teams building entire offensive systems around perimeter shooting. As a player who relies heavily on the three-point shot, I'm curious to examine whether this dramatic shift is effective. This study investigates how three-point shooting volume and efficiency relate to team success.

Research Questions:

Research Question 1 – Does the number of three-point attempts per game correlate with a team's win percentage?

Research Question 2 – Does a higher three-point shooting percentage lead to a higher win percentage?

Methods

Data Collection: The data was collected from Basketball Reference, consisting of a sample size of 150 observations spanning five NBA seasons (2019-2024). Each observation represents one NBA team's seasonal statistics. Sampling units are NBA teams.

Measures:

- Response Variable: Win percentage (ranging from 0 to 1).
- Numeric Explanatory Variable: Average three-point attempts per game.
- Categorical Explanatory Variable: Three-point efficiency (High = ≥36.65%, Low = <36.65%). High and low was based on league median for three-point shooting in the 2023-2024 season (36.65%).

Analysis Method: General Linear Model (GLM) was run in RStudio.

Descriptives

Table 1 – Descriptive Statistics

	Median	***IQR***
Win Percentage	0.524	0.208
	Median	***IQR***
Threes Attempted/Game	34.2005	5.012
	High Count	***Low Count***
Three-Point Efficiency	57	93

Results

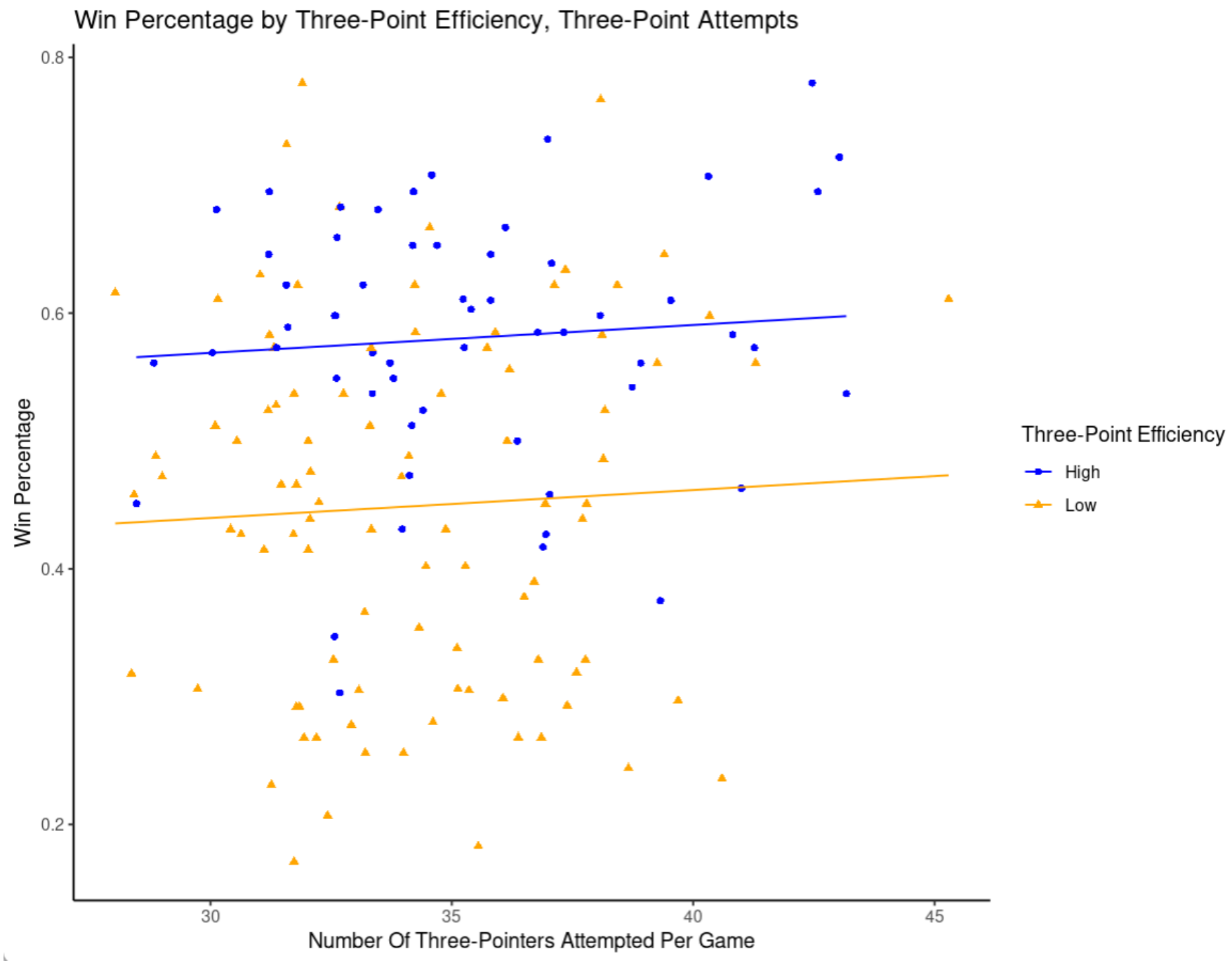
Results table:

Table 2 – Model Results

	Estimate	t-stat	St. Error	p-value
Intercept	0.503	4.571	0.110	1.02e-05
threes_efficiencyLow	-0.129	-5.903	0.022	2.37e-08
threes_attempted	0.002	0.709	0.003	0.479

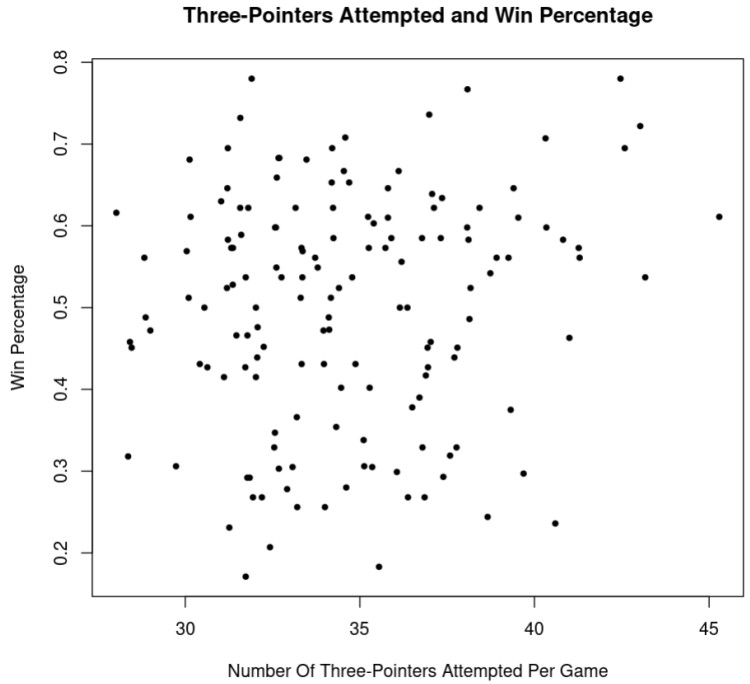
Model df: 147

Adjusted R-squared: 0.194

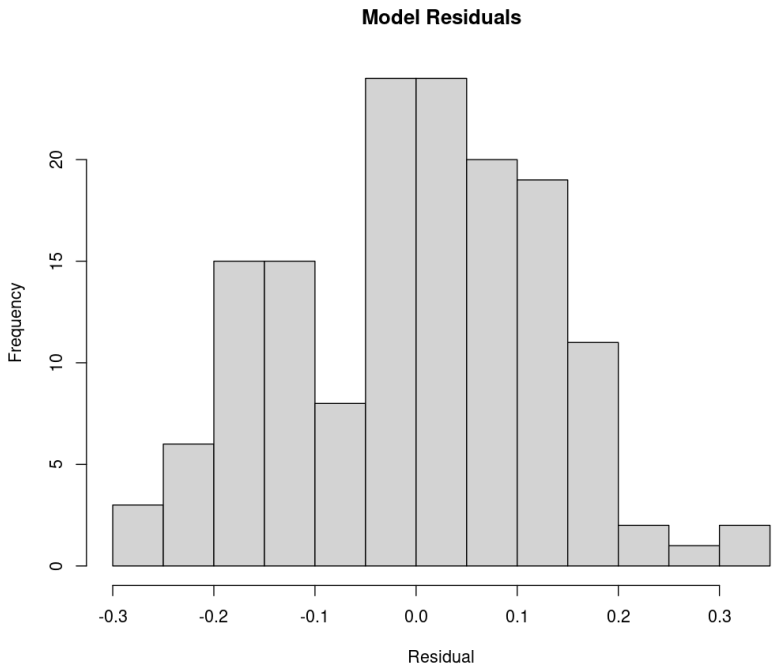

Win Percentage by Three-Point Efficiency, Three-Point Attempts
Win Percentage
0.8
0.6
0.4
0.2
30
35
40
45
Number Of Three-Pointers Attempted Per Game
Three-Point Efficiency
High
Low

Assumptions

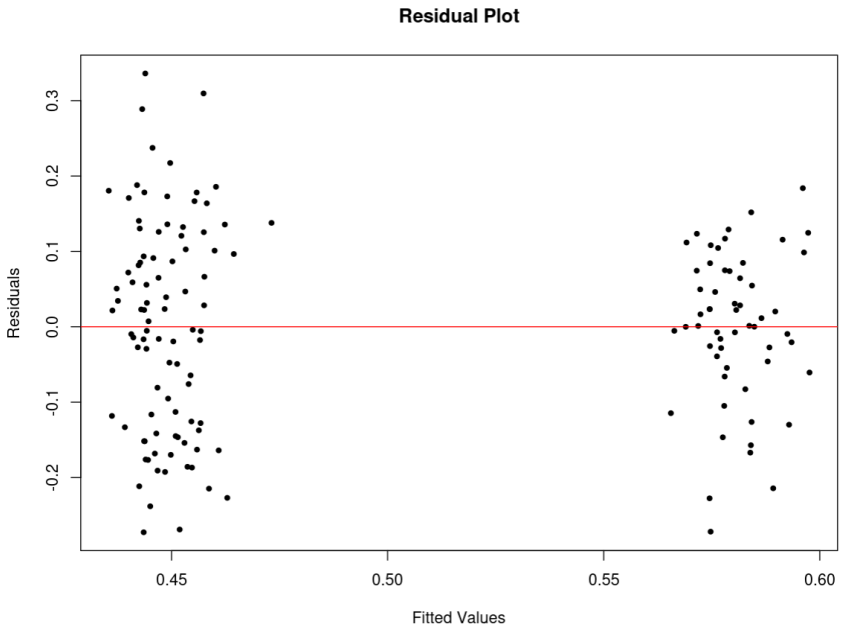
- **Random Sample:** The dataset was not randomly sampled as it included all NBA teams from the 2019-2024 seasons. Therefore, this may limit the ability to generalize about other eras.
- **Independent Observations:** Each observation represents a unique team-season combination. Since rosters and strategies change seasonally, the observations are independent.
- **Clear Non-Linear Relationship:** Based on the plot below, there is a clear non-linear relationship between the number of threes attempted per game and win percentage.



- **Normality:** Based on the histogram below, the population of values for win percentage are normally distributed for the number of threes attempted per game for each observation.



- **Variance:** Based on the plot below, idk


Residual Plot
Residuals
0.3
0.2
0.1
0.0
-0.1
-0.2
0.45
0.50
0.55
0.60
Fitted Values

Discussion

- Holding the number of attempts constant, three-point shooting efficiency is a significant linear predictor of win percentage (t=-5.903, df=147, p=2.37e-08).
  - Teams with low three-point shooting efficiency had a win percentage that was 0.129 lower than teams with high efficiency, on average.
- Holding three-point shooting efficiency constant, the number of threes attempted is not a significant linear predictor of win percentage (t=0.709, df=147, p=0.479).

Limitations:

- The dataset was not randomly sampled, as it included all NBA teams from the 2019-2024 seasons. This limits the ability to generalize finding to other eras or contexts.
- The analysis did not account for potential confounding variables such as turnovers, game pace, injuries, or mid-season trades, which could all influence win percentage.
- Team observations across different seasons may not be fully independent, as some organizations maintain consistent coaching or player cores.
- Variance?

Implications: Overall, teams should focus more on improving their three-point shooting efficiency rather than just increasing the number of attempts, if they want to win more in today's NBA.

Future Research: Next time, I'd include both three-point shooting and points in the paint to see which one helps teams win more. That would provide a better picture of which scoring style leads to more success. If I did this study again, I would include more NBA seasons to see how the relationship between the number of three-point attempts and winning has changed over time, which would help reveal if the recent trend of more attempts is helping or hurting.

References:

Grimes, P. J. (2024, October 30). The NBA has a 3-pointer problem. USA Today.

https://ftw.usatoday.com/2024/10/nba-three-pointers-record-attempts-problem

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